

COMPARISON OF VESTIBULAR MIGRAINE AND HEALTHY CONTROL GROUPS IN TERMS OF AUDIOLOGICAL, VESTIBULAR, AND OCULOMOTOR FINDINGS

20-Participant Pilot Statistical Analysis and Manuscript-Ready Academic Report

Report scope: descriptive statistics, group comparisons, categorical analyses, correlations,
hypothesis evaluation, and manuscript-ready results text

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EXECUTIVE SUMMARY

- The study included 20 participants: 10 patients diagnosed with vestibular migraine and 10 healthy controls.
- The DHI total score was markedly higher in the patient group than in the control group (median: 41.00 vs 10.00; $p < 0.001$).
- The OPK mean was lower in the patient group (median: 34.23 vs 72.73; $p < 0.001$). Low OPK was also more frequent categorically in the patient group (8/10 vs 0/10; $p < 0.001$).
- Saccade velocity was lower in the patient group (median: 264.78 vs 343.65; $p = 0.0376$).
- A significant difference was found in the horizontal vHIT mean; however, because the value was higher in the patient group, the prespecified hypothesis of “lower gain in the patient group” was not supported.
- No significant group differences were detected for vertical and overall vHIT, smooth pursuit, or saccade latency.
- No statistically significant correlations were found between DHI and objective test parameters in the patient group.
- Because comparable category data for the control group were unavailable for the audiometric finding variable, the between-group Fisher test should not be interpreted academically.
- The sample is small and multiple comparisons were performed. Therefore, the findings should be considered pilot and exploratory rather than confirmatory.

1. METHODS AND STATISTICAL APPROACH

Sample. Analyses were conducted on 20 individuals: 10 patients diagnosed with vestibular migraine and 10 healthy controls.

Descriptive statistics. Numerical variables were summarized using n, mean, standard deviation, median, minimum, and maximum. Because of the small sample, medians were emphasized in group comparisons.

Group comparisons. Independent patient and control groups were compared for numerical variables using a two-sided Mann-Whitney U test.

Categorical analyses. For binary variables, Fisher’s exact test was used when valid observations were available in both groups. If all category data were missing for one group, the test result was not interpreted.

Correlation analysis. Within the vestibular migraine group, associations between DHI total score and objective test parameters were evaluated using Spearman rank correlation.

Significance level. Statistical significance was defined as $p < 0.05$. However, because no multiple-comparison correction was applied, p-values should be interpreted in an exploratory framework.

Data principle. Missing values were not coded as normal findings; analyses were performed only on data present in the source file and convertible to numeric form.

Abbreviations: DHI, Dizziness Handicap Inventory; vHIT, video head impulse test; SP, smooth pursuit; OPK, optokinetic test; SD, standard deviation; OR, odds ratio.

2. RESULTS

Table 1. Descriptive statistics for the patient and control groups

Variable	Group	n	Mean	SD	Median	Minimum	Maximum
DHI total score	Patient	10	50.00	19.6638	41.00	30.00	84.00
DHI total score	Control	10	9.60	6.8508	10.00	0.00	22.00
Horizontal vHIT mean	Patient	10	0.9350	0.0767	0.9575	0.7450	1.0200
Horizontal vHIT mean	Control	10	0.8560	0.0689	0.8600	0.7350	0.9450
Vertical vHIT mean	Patient	10	0.8085	0.1342	0.8338	0.4525	0.9475
Vertical vHIT mean	Control	10	0.8382	0.0413	0.8500	0.7550	0.8950
Overall vHIT mean	Patient	10	0.8507	0.0986	0.8625	0.6033	0.9717
Overall vHIT mean	Control	10	0.8442	0.0413	0.8458	0.7633	0.9117
Smooth pursuit mean	Patient	10	80.860	8.0373	82.950	66.300	92.950
Smooth pursuit mean	Control	10	84.145	2.3326	83.375	81.000	88.650
OPK mean	Patient	10	35.300	23.1609	34.225	3.000	72.000
OPK mean	Control	10	71.980	3.2531	72.725	65.900	76.650
Saccade latency mean	Patient	10	202.000	30.7047	194.250	165.000	255.000
Saccade latency mean	Control	10	200.900	5.7581	200.750	191.500	213.000
Saccade velocity mean	Patient	10	277.665	65.6088	264.775	202.800	367.000
Saccade velocity mean	Control	10	344.740	17.5356	343.650	314.350	376.350

Note. Because of the small sample and the non-parametric comparison approach, medians should be prioritized when interpreting the findings.

Table 1 interpretation: The median DHI score in the patient group is approximately four times that of the control group. Median OPK and saccade velocity values are lower in the patient group. In contrast, overall vHIT and smooth pursuit medians are similar between groups. The wider dispersion observed in the patient group for OPK, vertical vHIT, and saccade velocity may indicate greater inter-individual variability than in the control group; however, this interpretation should be made cautiously because of the small sample.

Table 2. Mann-Whitney U test results for numerical variables

Variable	Patient n	Control n	Patient median	Control median	U	p	Result
DHI total score	10	10	41.0000	10.0000	100.0	0.0002	Significant difference
Horizontal vHIT mean	10	10	0.9575	0.8600	82.5	0.0155	Significant difference*
Vertical vHIT mean	10	10	0.8338	0.8500	46.5	0.8205	Not significant
Overall vHIT mean	10	10	0.8625	0.8458	63.0	0.3443	Not significant
Smooth pursuit mean	10	10	82.9500	83.3750	41.0	0.5205	Not significant
OPK mean	10	10	34.2250	72.7250	4.0	0.0006	Significant difference
Saccade latency mean	10	10	194.2500	200.7500	36.5	0.3256	Not significant
Saccade velocity mean	10	10	264.7750	343.6500	22.0	0.0376	Significant difference

Note. A two-sided test was used. *The horizontal vHIT difference is statistically significant; however, because the patient median is higher, the direction is opposite to the prespecified low-gain hypothesis.

Table 2 interpretation: Significant differences are present between the patient and control groups for DHI total score, OPK mean, and saccade velocity. DHI is higher in the patient group, whereas OPK and saccade velocity are lower. Horizontal vHIT also differs significantly; however, because the value is higher in the patient group, this finding does not support the hypothesis of peripheral vestibular hypofunction. No significant differences were detected for vertical/overall vHIT, smooth pursuit, or saccade latency. In particular, the $p=0.0376$ result for saccade velocity should be viewed as borderline and exploratory in light of multiple testing and the small sample.

Table 3. Fisher's exact test results for categorical variables

Variable	Patient: Yes	Patient: No	Control: Yes	Control: No	OR	p	Result
Low vHIT gain	1	9	2	8	0.4444	1.0000	No significant association
Low OPK	8	2	0	10	∞	0.0007	Significant association
Prolonged saccade latency	2	8	0	10	∞	0.4737	No significant association
Audiometric finding	5	5	No data	No data	-	-	Not comparable
Abnormal tympanometry	0	10	0	10	Undefined	1.0000	No events in either group

Note. Low vHIT gain was classified as <0.80 ; low OPK as <56 ; and prolonged saccade latency as >250 ms. Because of zero cells, some OR values are infinite or undefined. The test result for the audiometric finding was not reported because valid category data were unavailable in the control group.

Table 3 interpretation: Low OPK was observed in 8/10 participants in the patient group and in none of the controls, and this distribution was statistically significant. No significant association was found for low vHIT gain or prolonged saccade latency. The absence of abnormal tympanometry in both groups indicates that no events occurred in this sample; a p-value of 1.0000 does not prove group equivalence. Because category data for the control group were unavailable for the audiometric finding, the former 5/10 versus 0/0 presentation is not academically valid and no between-group conclusion can be drawn for this variable.

Table 4. Spearman correlations between DHI and objective test parameters in the vestibular migraine group

Variable	n	Spearman rho	p	Direction	Result
Horizontal vHIT mean	10	0.2012	0.5772	Positive	Not significant
Vertical vHIT mean	10	-0.0915	0.8016	Negative	Not significant
Overall vHIT mean	10	0.0826	0.8206	Positive	Not significant
Smooth pursuit mean	10	0.1707	0.6372	Positive	Not significant
OPK mean	10	-0.4098	0.2396	Negative	Not significant
Saccade latency mean	10	0.1768	0.6250	Positive	Not significant
Saccade velocity mean	10	-0.2317	0.5195	Negative	Not significant

Note. Correlation analyses were performed only on the 10 participants in the patient group. rho: Spearman rank correlation coefficient.

Table 4 interpretation: None of the examined correlations reached $p < 0.05$. The largest absolute coefficient was between DHI and OPK ($\rho = -0.4098$); however, with $p = 0.2396$, the statistical evidence is insufficient in this sample. Correlation analyses with $n = 10$ have limited power; therefore, the appropriate wording is “no significant association could be demonstrated” rather than “there is no association.”

Table 5. Evaluation of hypotheses using the statistical findings

Hypothesis	Key finding	p	Decision	Academic interpretation
H1g: DHI is higher in the patient group	41.00 vs 10.00	0.0002	Supported	The difference is in the expected higher-value direction.
H1a: Horizontal vHIT gain is lower in the patient group	0.9575 vs 0.8600	0.0155	Not supported	The significant difference is in the opposite direction; the patient median is higher.
H1b: OPK is lower in the patient group	34.225 vs 72.725	0.0006	Supported	The difference is in the expected lower-value direction.
H1c: Saccade latency is longer in the patient group	194.25 vs 200.75	0.3256	Not supported	No significant difference could be demonstrated.
Saccade velocity differs between groups	264.775 vs 343.650	0.0376	Supported	A lower value was observed in the patient group.
Smooth pursuit differs between groups	82.950 vs 83.375	0.5205	Not supported	No significant difference could be demonstrated.

Note. “Supported” refers only to the statistical direction and significance observed in this pilot sample; it does not imply causality or clinical validation.

Table 5 interpretation: The directional hypotheses for DHI and OPK were supported. Although a significant difference was found for horizontal vHIT, the direction of the difference was opposite to the hypothesis. The between-group difference in saccade velocity was supported, whereas the smooth pursuit and saccade latency hypotheses were not supported. Hypothesis decisions should be regarded as preliminary findings in the context of multiple comparisons and the limited sample.

3. MANUSCRIPT-READY RESULTS TEXT

The study included a total of 20 participants: 10 patients diagnosed with vestibular migraine and 10 healthy controls. For all numerical variables examined, usable data were available for 10 participants in each group. Because of the small sample size, group comparisons were performed using the Mann-Whitney U test and results were interpreted primarily using median values.

The DHI total score was significantly higher in the vestibular migraine group than in the control group. The median DHI score was 41.00 in the patient group and 10.00 in the control group ($U=100.0$; $p<0.001$). This finding indicates that the subjective impact of dizziness was more pronounced in the patient group.

The horizontal vHIT mean differed significantly between groups. The patient median was 0.9575 and the control median was 0.8600 ($U=82.5$; $p=0.0155$). However, the direction of the difference was opposite to the hypothesis predicting lower gain in the patient group. Vertical vHIT medians were 0.8338 and 0.8500 in the patient and control groups, respectively ($p=0.8205$), while overall vHIT medians were 0.8625 and 0.8458 ($p=0.3443$); no significant differences were detected for these two variables.

The median smooth pursuit mean was 82.950 in the patient group and 83.375 in the control group, with no significant between-group difference ($U=41.0$; $p=0.5205$). In contrast, the OPK mean was markedly lower in the patient group. The median OPK was 34.225 in the patient group and 72.725 in the control group ($U=4.0$; $p<0.001$). In the categorical evaluation, low OPK was observed in 8/10 patients and 0/10 controls, and the distribution was significant (Fisher's exact test, $p<0.001$).

The median saccade latency was 194.25 ms in the patient group and 200.75 ms in the control group, and no significant difference could be demonstrated ($U=36.5$; $p=0.3256$). The median saccade velocity was 264.775 in the patient group and 343.650 in the control group, with a significantly lower value in the patient group ($U=22.0$; $p=0.0376$). This finding should be considered exploratory because of multiple testing and the small sample.

Low vHIT gain was observed in 1/10 patients and 2/10 controls, with no significant association between groups ($p=1.0000$). Prolonged saccade latency was observed in 2/10 patients and 0/10 controls; the difference was not significant ($p=0.4737$). No abnormal tympanometry findings were detected in either group. Because valid category data for the control group were unavailable for the audiometric finding variable, a between-group comparison could not be performed for this variable.

Within the vestibular migraine group, no statistically significant correlations were found between DHI total score and horizontal, vertical, or overall vHIT, smooth pursuit, OPK, saccade latency, or saccade velocity. The largest absolute correlation was a negative association between OPK and DHI, but it was not significant ($\rho=-0.4098$; $p=0.2396$).

4. DISCUSSION

In this pilot analysis, the most prominent findings distinguishing the vestibular migraine group from healthy controls were a higher DHI total score and lower OPK and saccade velocity values. The DHI finding indicates a more pronounced subjective dizziness burden in the vestibular migraine group than in the control group. However, it should be recognized that DHI assesses subjective impact, whereas vestibular and oculomotor tests assess physiological responses at the time of measurement.

The fact that the lower OPK mean was significant both as a continuous variable and at the categorical classification level makes it one of the most consistent objective findings in this dataset. Nevertheless, the wide distribution of OPK values in the patient group and the fact that the sample consisted of only 10 patients warrant caution regarding effect magnitude and generalizability. The infinite OR produced by the zero cell in the categorical analysis reflects the absence of events in the control group, not a precise estimate of effect magnitude.

The lower saccade velocity in the patient group may indicate a difference in oculomotor performance. In contrast, the lack of a significant difference in saccade latency suggests that not all components of the saccadic response are affected in the same direction. Because the p-value for saccade velocity is close to the 0.05 threshold and no multiple-comparison correction was applied, this result should be considered a preliminary finding requiring testing in new samples rather than a confirmatory result.

The significant difference in the horizontal vHIT mean was in the direction of higher gain in the patient group. Therefore, the result does not support the hypothesis predicting low gain due to peripheral vestibular hypofunction in the patient group. The non-significant vertical and overall vHIT findings also indicate that there is no consistent low-gain pattern across vHIT measures.

Smooth pursuit values were similar between groups. Thus, the current sample does not suggest a common impairment across all oculomotor parameters; the differences in OPK and saccade velocity may be more selective. However, measurement conditions, device characteristics, and test reproducibility were not separately examined in this report.

The inability to demonstrate significant correlations between DHI and objective test parameters suggests that subjective symptom burden and test results did not show a linear or monotonic parallel pattern in this small sample. However, because the correlation analyses were based on only 10 patients, the power to detect moderate associations is limited.

The absence of abnormal tympanometry findings in both groups indicates that no differences were observed in middle-ear measurements in this sample. However, zero events in both groups do not constitute evidence of equivalence. The absence of control data for the audiometric finding variable prevents evaluation of this variable between groups; no comparative audiometric conclusion should be written until the control data are completed.

5. LIMITATIONS AND REPORTING NOTES

- The sample size is limited to 10 participants per group. Statistical power is therefore low, and medians may be influenced by individual observations.
- Multiple numerical and categorical comparisons were performed, but no multiple-comparison correction was applied in this report. Findings with p-values close to 0.05 should be confirmed in an independent sample.
- Effect sizes and confidence intervals are not included in the current output. Before manuscript submission, appropriate effect sizes and, where possible, confidence intervals should be added for Mann-Whitney comparisons.
- Control-group category data are unavailable for the audiometric finding variable. Fisher's exact test or between-group audiometric interpretation should not be performed until this missing information is completed.
- The clinical rationale for categorical thresholds should be stated explicitly in the Methods section, and it should be verified whether these thresholds were defined before the study began.
- Statistical significance is not equivalent to clinical importance. Findings should be interpreted together with clinical assessment, measurement reliability, and larger samples.

6. CONCLUSION

In this 20-participant pilot analysis, the DHI total score was higher and the OPK mean and saccade velocity were lower in the vestibular migraine group. Low OPK was also more frequent in the patient group in the categorical analysis. Although a significant difference was found in the horizontal vHIT mean, the direction was opposite to the low-gain hypothesis; no significant differences could be demonstrated for vertical and overall vHIT, smooth pursuit, or saccade latency. No significant correlations were detected between DHI and objective test parameters. The findings are pilot and exploratory and should be confirmed with a larger sample, complete audiometric data, effect-size reporting, and control for multiple comparisons.

APPENDIX 1. TECHNICAL DATA CHECK

For all numerical analysis variables, 10 usable observations are available in both the patient and control groups. This check covers numerical variables only; the lack of control-group data in the audiometric finding category is evaluated separately.

Technical warning: Although there are no missing observations for the numerical variables, category data for the audiometric finding variable are unavailable in the control group. Therefore, rather than stating that “no warnings were identified,” the report explicitly records that the audiometric comparison could not be performed.

Appendix Table 1. Number of usable observations for numerical variables

Variable	Patient usable	Patient missing	Control usable	Control missing
DHI total score	10	0	10	0
Horizontal vHIT	10	0	10	0
Vertical vHIT	10	0	10	0
Overall vHIT	10	0	10	0
Smooth pursuit	10	0	10	0
OPK	10	0	10	0
Saccade latency	10	0	10	0
Saccade velocity	10	0	10	0

Note. The control-group data missing for the categorical audiometry variable are not included in this table.

Appendix Table 2. Source columns used in the analysis

Analysis	Source columns
DHI total	dhi_toplam
vHIT horizontal	vHIT_HOR_Sag_Gain; vHIT_HOR_Sol_Gain
vHIT vertical	vHIT_right_Posterior_Gain; vHIT_Left_Posterior_Gain; vHIT_Left_Anterior_Gain; vHIT_Right_Anterior_Gain
Smooth pursuit	SP_Sag_Goz_Kazanc_% ; SP_Sol_Goz_Kazanc_%
OPK	OPK_Sol45_Kazanc_%; OPK_Sag45_Kazanc_%
Saccade latency	Sakkad_Sol_Latans_ms ; Sakkad_Sag_Latans_ms
Saccade velocity	Sakkad_Sol_Velocity; Sakkad_Sag_Velocity
Audiometric finding	Not
Tympanometry result	Timpanometri

Note. Column names are shown using the technical names in the source Excel file.